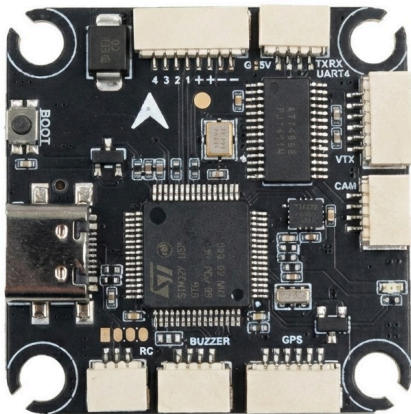


User Guide



AF-F4 T10 nano Overview

AF-F4 nano + AG-F4-1A Gimbal · ArduPilot

AF-F4 T10 nano Specifications

• Basic Specifications

MCU	STM32F405
IMU	ICM-42688-P
Barometer	SPL06 (current novaX hardware definition)
Gimbal	AG-F4-1A (MNT_TYPE 13)
Operating Voltage	9 – 25 V DC
USB Input	USB Type-C (Firmware & Power)
PWM Output	8 motor/servo channels + 1 LED channel
Serial Ports	6 hardware UARTs + USB configured; verify connector pinout

AF-F4 T10 nano Specifications

• Basic Specifications

Serial Mapping	SERIAL1 = USART1 · SERIAL2 = USART2 · SERIAL3 = USART3 · SERIAL4 = UART4 · SERIAL5 = UART5 · SERIAL6 = USART6 (USB = SERIAL0)
RC Input	PWM / PPM / S.Bus
Size	39.4 × 39.4 mm
Weight	9.3 g
Supported F/W	ArduPilot

AF-F4 T10 nano Pinout

• Pinout

AF-F4 nano Pinout

Both sides were straight on. Each table lists the pins left → right (side connectors: top → bottom) as you see them in the photo. The board silks names the ESC, GLRS and every bottom pad; the other top connectors keep the maker's order. There are no pin numbers.

red = power black = GND blue = signal grey = not connected

Top side

ESC					GLRS				
8-Pin: USB, 4: 3.3V, 5: GND, 6: 5V, 7: TX, 8: RX					8-Pin: UART1: 6: 5V, 7: TX, 8: RX				
54	53	52	51	+	GND	5V	TX	RX	



VTX	
6-Pin: UART1 TX: maker's order	
NC	
NC	
NC	
VTX	
TX	

CAM	
8-Pin: video in: maker's order	
5V	
GND	
CAM	

RC			GPS				
8-Pin: PP3: maker's order			8-Pin: UART1 + GPRS: maker's order				
GND	5V	PP3	SDA1	SCL1	RST	TX	5V GND

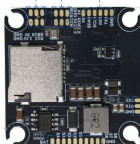
BUZZER			
8-Pin: UART1 RX: maker's order			
5V	RX3	BUZZ	GND

Bottom side (pads)

RGB pads		
UART1 / UART2: LED or serial		
RGB	GND	5V
RX3	TX3	RX3

SERIAL pads		
UART1 RX: RC input		
GND	5V	SERIAL
GND	3.3V	RX2

DC / UART pads					
UART1					
5V	SDA1	SCL1	RST	TX	GND



VTX pads	
UART1 TX: VTX control	
TX	
VTX	
GND	

MOTOR / POWER pads							
Motor signals GND → battery							
RED	54	53	52	51	5V	5V	GND

Also on the board: M3U STM32F405RG (168 MHz; 1 MB flash; 192 KB RAM); IMU ICM-42055-P; OSD AT79SG; BOOT button and USB Type-C on the top; TF card slot and the BEE on the bottom.

AF-F4 T10 nano Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set AHRS_ORIENTATION.
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 9 – 25 V DC.
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

AF-F4 T10 nano Firmware

• Available Firmware

Type	Version	Date
ArduPilot (.apj package)	1.3.4	2026-07-11
Bootloader + App (merged HEX · DFU / SWD)	1.3.4	2026-07-11

Download files from the product page (QR).

AF-F4 T10 nano Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.



<Detailed Manual>

